

Hotel Booking System

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Algorithms:

1. Start
2. Input the number of guests, N .
3. While $N < 1$, ask for N again.
4. set Guest $\leftarrow 1$ and HotelTotal $\leftarrow 0$.
5. While Guest $\leq N$
 1. Input Season.
 2. While Season is neither Peak nor off-Peak,
ask again.
 3. Input Room.
 4. While room is neither Standard, Deluxe, nor Suite, ask
again.
 5. Input Nights.
 6. While Nights < 1 , ask again.
 7. If season is peak:
 - If room is Standard, set rate to 5000.
 - Otherwise, if room is Deluxe, set rate to 8000.
 - Otherwise, set rate to 12000.
 8. Otherwise:
 - If room is Standard, set rate to 3000.
 - Otherwise, if room is deluxe, set rate to 5000.
 - Otherwise, set rate to 8000.
 9. Calculate Amount $\leftarrow$ Rate $\times$ Nights.
 10. If Nights > 7 , calculate Discount $\leftarrow$ Amount $\times 15/100$.
 11. Otherwise, set Discount $\leftarrow 0$.
 12. Calculate Final price $\leftarrow$ Amount - Discount.
 13. Display the guest number and final price.
 14. Add Final price to HotelTotal.
 15. Increase Guest by 1.
6. Display HotelTotal.
7. Stop.

Pseudo code.

Begin

Let N , be the user input for number of guests.

While ($N < 1$), do

 Print "Enter at least one guest"

 Input N

END while

Let Guest $\leftarrow 1$, HotelTotal $\leftarrow 0$

While (Guest $\leq N$), do

 Input Season

 While (Season $\neq$ "Peak" AND Season $\neq$ "Off-Peak"), do

 Print "Enter Peak or Off-Peak"

 Input Season

 End While.

 Input Room.

 While (Room $\neq$ Standard AND Room $\neq$ Deluxe AND Room $\neq$ Suite)

 Print "Enter Standard, Deluxe or Suite"

 Input Room

 END While

 Input Nights

 While (Nights < 1), do

 Print "Enter at least one night"

 Input Nights

 End while

 If (Season = "Peak"), then

 If (Room = "Standard"), then

 Rate $\leftarrow 5000$

 Else-If (Room = "Deluxe"), then

 Rate $\leftarrow 8000$

Else

Rate $\leftarrow$ 12000

END IF

Else

IF (Room = "Standard"), then

Rate $\leftarrow$ 3000

Else IF (Room = "Deluxe"), then

Rate $\leftarrow$ 5000

Else

Rate $\leftarrow$ 8000

END IF

END IF

Amount $\leftarrow$ Rate * Nights

IF (Nights > 7), then

Discount $\leftarrow$ Amount * 15 / 100

Else

Discount $\leftarrow$ 0

END IF

FinalPrice $\leftarrow$ Amount - Discount

HotelTotal $\leftarrow$ HotelTotal + FinalPrice

Print "Guest number: ", Guest

Print "Final price: ", FinalPrice

Guest $\leftarrow$ Guest + 1

END while

Print "Hotel total: ", HotelTotal

STOP

Flowchart





